

Software Engineering Assignment 3

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Section: F

System: Online Food Delivery System

Architecture Selection: We chose Microservices Architecture for the Online Food Delivery System.

Why this architecture

Order placement and restaurant browsing get a lot more traffic than something like sending notifications, especially around lunch and dinner time. With microservices, we can scale up just the Order and Restaurant services during busy hours instead of scaling the whole system, which saves on cost and keeps things running smoothly.

The system also depends on a few external services - the payment gateway, SMS/push notifications, and delivery partner tracking. If the notification service or delivery tracking goes down for some reason, customers can still browse restaurants, place orders and pay, since each service runs on its own. In a more tightly coupled setup, one failing part could easily bring the whole system down.

Security

Since each service is deployed separately, the Payment Service can be locked down with stricter access controls and encryption than something like the Restaurant Service, which handles less sensitive data. This way, if a lower-priority service gets compromised, it doesn't automatically put payment data at risk, because the services don't share a database.

Performance

Because the Order Service is separate from the rest, it can have its own dedicated database and caching tuned for fast order writes, without waiting on heavier services like delivery tracking. This keeps order placement fast even when other parts of the system are under load or being updated.

Components identified

Order Service, Payment Service, Restaurant Service, Delivery Tracking Service, Notification Service. Order Service talks to Payment (for authorizing payments), Restaurant (for menu and availability), Delivery (to dispatch orders) and Notification (for order status updates).

Component Diagram

